UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Important points

Definitions, statements, key points, "why" reasoning and exam traps, topic by topic.

Class 002 · Geometrical interpretation of partial derivatives · compiled 14 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S002·7	Deck 002, slide 7 (rendered .pptx)
T002 00:06:10	AV-summary transcript 002, segment start time
V002 00:07:45	Frame grabbed from the class 002 video
CW U1 P2 · ESA Dec-23 1a	Unit 1 class-work problem 2 · past-paper question

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1 Higher-order partial derivatives

Definitions

First-order partial derivative. The partial derivative $\partial f/\partial x$ obtained by differentiating once (here, with respect to x while y is held constant). S002·3

Second-order partial derivatives. $\dfrac{\partial^2 f}{\partial x^2}$, $\dfrac{\partial^2 f}{\partial y^2}$, $\dfrac{\partial^2 f}{\partial x \partial y}$, $\dfrac{\partial^2 f}{\partial y \partial x}$, which are obtained by differentiating twice. S002·3

Higher-order partial derivatives. Third, fourth and nth order derivatives involve 3, 4 or n times differentiation respectively. S002·3

Mixed partial derivative. A higher-order partial in which you differentiate with respect to more than one variable, e.g. f_{xy} and f_{yx}. S002·9 · T002 00:00:06

Statements in the class material

> *"A function of 2 variables has two first order derivatives, four second order derivatives and 2^n of nth order derivatives. A function of m independent variables will have m^n derivatives of order n."* S002·3

Key points

1. A partial derivative is taken with respect to one variable while every other variable is held constant. ESA ★★ T002 00:00:06
2. Differentiating a first-order partial again gives a second-order partial; n differentiations give an nth-order partial. ESA ★★ S002·3
3. For $f(x, y)$: 2 first-order (f_x, f_y), 4 second-order ($f_{xx}, f_{yy}, f_{xy}, f_{yx}$), 2^n of order n. ESA ★★ S002·3 · T002 00:00:44
4. For $f(x_1, \ldots, x_m)$: m^n derivatives of order n; e.g. m^2 second-order partials. ESA ★ S002·3 · T002 00:02:13
5. The two mixed partials "were equal" in the class examples: $f_{xy} = f_{yx}$ for the smooth functions of this unit. ESA ★★ (transcript only) T002 00:00:24

Exam traps

Exam trap. In f_{xy} the order is x first, then y. Some books write the same derivative as $\partial^2 f / \partial y \, \partial x$. Both mean the same, and for this unit's functions the order does not change the answer. (added for clarity)

2 The one-variable picture: derivative as slope

Definitions

Derivative (one variable). For $y = f(x)$, dy/dx is the change in y produced by a change in position along the x-axis; at a point it equals the slope of the tangent to the curve. S002·4 · T002 00:02:50

Statements in the class material

> *"We know that the derivative of a single valued function gives the slope of the tangent line to the curve at the given point."* S002·4

Key points

1. A derivative always represents change: dy/dx is the change on the curve as you change your position on the x-axis. ESA ★ T002 00:02:50 · T002 00:02:59
2. At a particular point, the derivative is the slope of the tangent at that point. ESA ★ S002·4 · T002 00:03:13

3 A function of two variables is a surface

Definitions

Bivariate function. A function $z = f(x, y)$ of two independent variables. Its graph is a surface in three dimensions. T002 00:04:16

Curve on a surface. If the points (x, y) are restricted to a curve in the xy-plane (circle, parabola, ...), the points $(x, y, f(x, y))$ form a curve that lies on the surface. S002·5

Statements in the class material

> *"If the points (x,y) lie on a circle in the xy plane, then z = f(x,y) is a curve which looks like the graph shown. If the points (x,y) lie on a parabola in the xy plane, then z = f(x,y) is a curve which looks like the graph shown."* S002·5

Key points

1. A function of two independent variables always gives a surface in three dimensions: each (x, y) has one z. ESA ★ T002 00:04:16
2. Restricting (x, y) to a curve in the xy-plane (circle, parabola) picks out a curve on the surface. ESA ★ S002·5 · T002 00:03:32

4 First-order partials as slopes of tangents

Definitions

Partial derivative, geometrically. The slope of the tangent line to the surface $z = f(x, y)$ at (x_0, y_0), along the direction of the variable you differentiate with respect to. S002·6

Trace (plane section). The curve cut from the surface by a vertical plane such as $y = y_0$ (giving $z = f(x, y_0)$) or $x = x_0$ (giving $z = f(x_0, y)$). ADDED FOR CLARITY S002·7

Statements in the class material

> *"Partial derivative of the function f(x,y) at the point (x_0, y_0) can be visualized as the slope of the tangent line along the direction of the independent variable with respect to which we would differentiate the function."* S002·6

Key points

1. A surface has infinitely many tangents at a given point, one for every direction. ESA ★★ T002 00:05:24 · T002 00:05:35
2. z_x at (x_0, y_0) is the slope of the tangent to the surface at that point which is parallel to the x-axis. ESA ★★ T002 00:06:10 · T002 00:06:27
3. z_y at (x_0, y_0) is the slope of the tangent parallel to the y-axis. ESA ★★ T002 00:07:10 · T002 00:07:54
4. To find $\partial f/\partial x$ at (x_0, y_0): put $y = y_0$, which gives a curve $f(x, y_0)$ in one variable, then differentiate it at $x = x_0$. ESA ★★ S002·7
5. The cutting plane $y = y_0$ slices the surface in the curve whose slope is f_x; the plane $x = x_0$ gives the curve whose slope is f_y. ESA ★★ S002·7 · V002 00:07:45

Why?

Why are there infinitely many tangents at one point of a surface?
From the point you can move in infinitely many directions in the xy-plane. Each direction traces a different curve on the surface, and each curve has its own tangent at P.
T002 00:04:36 · T002 00:05:24

Why does fixing $y = y_0$ turn $\partial f/\partial x$ into an ordinary derivative?
Once $y = y_0$, $f(x, y_0)$ depends on x alone. It is a plane curve in the plane $y = y_0$, and its ordinary derivative at x_0 is the slope of its tangent.
S002·7

Exam traps

Exam trap. "Parallel to the x-axis" describes the tangent's direction seen from above (it lies in the plane $y = y_0$). It does not mean the tangent is horizontal: its slope is $f_x(x_0, y_0)$, which is usually not zero. ADDED FOR CLARITY

Exam trap. Hold the other variable constant before differentiating. Do not substitute $x = x_0$ before differentiating with respect to x. ADDED FOR CLARITY

5 Slope in any other direction: the gradient

Definitions

Gradient. The vector $\nabla f = f_x\,\mathbf{i} + f_y\,\mathbf{j}$: the resultant of the two slope components along the x- and y-axes. S002·8 · T002 00:09:23

Directional derivative. The slope of the surface in the direction of a unit vector $\hat{u}$: $D_{\hat{u}} f = \nabla f \cdot \hat{u}$. It is studied in vector calculus. ADDED FOR CLARITY T002 00:09:54

Statements in the class material

> *"Slope of tangent along any arbitrary direction (Gradient of a function). What about other directions than parallel to x and y?"* S002·8

Key points

1. f_x and f_y only give slopes in two directions; most tangents are parallel to neither axis. ESA ★ S002·8 · T002 00:08:22
2. Any line in the plane can be resolved into components along the x- and y-axes; vector addition of the components gives the resultant. ESA ★ T002 00:09:06 · T002 00:09:23
3. The resultant of $f_x\,\mathbf{i}$ and $f_y\,\mathbf{j}$ is the gradient ∇f, studied in vector calculus. ESA ★ S002·8 · T002 00:09:44
4. The slope in a unit direction $\hat{u} = (a, b)$ is $af_x + bf_y$. It is greatest, equal to $|\nabla f|$, in the direction of ∇f. ESA ★ ADDED FOR CLARITY S002·8

Why?

Why can't f_x or f_y alone give the slope along a diagonal direction?
Each is the slope along one axis only. A diagonal direction has both an x- and a y-component, so both slopes contribute. Combining them gives the gradient, and its dot product with the unit direction gives the slope.
S002·8 · T002 00:09:06

6 Second-order partials: concavity and mixed partials

Definitions

Concave up in the x direction. The trace in the x direction bends upward; the x-slope increases as x increases. Indicated by $f_{xx} > 0$. S002·9

Mixed partial f_{xy}. Tells how the rate of change of $f(x, y)$ in the x direction is changing as we move in the y direction. S002·9

Statements in the class material

> *"If $f_{xx} > 0$ then f(x,y) is concave up in the x direction. If $f_{yy} > 0$ then f(x,y) is concave up in the y direction. Mixed partials tell us how a partial in one variable is changing in the direction of the other. f_{xy} tells us how the rate of change of f(x,y) in the x direction is changing as we move in the y direction."* S002·9

Key points

1. As in one variable (concave up or down from d^2y/dx^2), the second-order partials describe how the surface bends. ESA ★★ T002 00:11:03 · T002 00:11:19
2. $f_{xx} > 0$ means concave up in the x direction; $f_{yy} > 0$ means concave up in the y direction. ESA ★★ S002·9 · T002 00:11:38
3. $f_{xx} < 0$ (or $f_{yy} < 0$) means concave down in that direction. One surface can be up in x and down in y, as in a saddle. ESA ★★ added for clarity S002·9
4. f_{xy} is the change in f_x along the y direction; f_{yx} is the change in f_y along the x direction. ESA ★★ S002·9 · T002 00:11:45 · T002 00:12:04
5. The class demo, "Second-Order Partial Derivatives", lets you set the signs of f_{xx}, f_{yy} and f_{xy} and watch the red tangent line move in 3D and contour views. ESA ★ S002·10 · V002 00:12:40 · T002 00:12:40

Why?

Why does $f_{xx} > 0$ mean concave up in x?
f_{xx} is the rate of change of the x-slope as x increases. If it is positive, the tangent keeps tilting upward as you move along x, so the trace bends upward.
S002·9 · T002 00:11:19

Why do the 4 second-order partials usually give only 3 different values?
For the smooth functions used in this unit the mixed partials are equal, $f_{xy} = f_{yx}$ (Clairaut), leaving only f_{xx}, f_{yy} and f_{xy}.
added for clarity T002 00:00:24

7 Problems on partial derivatives

Definitions

Clairaut's theorem. If f_{xy} and f_{yx} are continuous near a point, they are equal there. The class plans to verify it in the next session. TRANSCRIPT ONLY ADDED FOR CLARITY T002 00:14:59

Key points

1. Evaluate at the point only after all the differentiations are done. ESA ★★ ADDED FOR CLARITY CW U1 P1 · ESA Dec-23 1a
2. To verify $f_{xy} = f_{yx}$, work out each side separately and compare them. ESA ★★ CW U1 P2

Exam traps

Exam trap. In these problems $\log$ is the natural logarithm: $\frac{d}{dy}\log y = \frac{1}{y}$. ADDED FOR CLARITY

Exam trap. For x^y: with respect to x it is a power, $y\,x^{y-1}$; with respect to y it is an exponential, $x^y \log x$. ADDED FOR CLARITY

★ Past-paper questions

Eight past ESA papers were checked: UE22MA141A Dec 2023, May 2024 and Jul 2023, and UE23MA141A Jan sets 1–4 and June 2024. None asks for the geometrical interpretation directly. Two 6-mark Q1 parts test higher-order partials, the computational half of this session. Other Unit 1 questions (Euler, Taylor, maxima, Lagrange, chain rule) belong to later sessions and are left out.

7 · Problems on partial derivatives

ESA Dec 2023 · 6 marks · UE22MA141A Q1.a · solved in W7.4

If $f(x,y) = e^x \log y + \cos y \log x$, find f_{xxx}, f_{xyy} and f_{yyy} at $(1, \pi/2)$.

ESA Jan 2024 · 6 marks · UE23MA141A set 4, Q1.c · solved in W7.5

If $\theta = t^n e^{-r^2/4t}$, what value of n will make $\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial \theta}{\partial r}\right) = \frac{\partial \theta}{\partial t}$?